Quantitative Evaluation of the Effects of Botulinum Toxin Injection Into the Temporalis Muscles and Parotid Glands

Botulinum toxin A (BTX-A) injection is a widely performed cosmetic procedure to improve the facial profile. In particular, BTX-A injection to reduce the temporalis muscles and parotid glands would be more beneficial for Asians because their faces appear wider and flatter than those of Westerners. Although BTX-A injection into the temporalis muscles has been used mainly for the treatment of headaches, myofascial pain, and bruxism,[1] it may have a cosmetic effect by reducing the volume of the muscles. To reduce the width of the lower face and make the chin slender, BTX-A injection into the masseter muscles is frequently performed.[2] However, in addition to the masseter muscles, the parotid glands are located at the mandibular angle; thus, reducing the volume of the parotid glands by BTX-A injection can also help make the jawline sharper.[3]

Here, we report a patient who received prabotulinum toxin Type A (PRA) (Nabota; Daewoong Pharmaceutical, Seoul, South Korea) injection into the temporalis muscles and parotid glands and another patient who received PRA into the temporalis muscle only. The changes in the volume of the temporalis muscles and parotid glands over time after PRA injection were measured by computed tomography (CT). This allowed for the quantitative evaluation of both volume reduction and effect duration after PRA injection. We found that the changes in the muscles and glands after toxin injection were different.



Figure 1. Sites of prabotulinum toxin Type A injection into the temporalis muscles (left) and parotid glands (right). Four units × eight points: 31G insulin syringe with 8-mm needle, vertical injection.

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Patient 1

A healthy 36-year-old woman visited our clinic to reduce the contour of her broad face. She received 32 units of PRA per parotid gland and temporalis muscle, respectively (128 units in total). The parotid gland is located in the preauricular area, behind the masseter muscle. When the patient bites, the posterior border of the masseter muscle can be palpated, making it easier to isolate the parotid gland. We used a 31G insulin syringe with an 8-mm needle. The injection sites are shown in Figure 1. Serial CT measurements were performed to determine the volume of the parotid glands and temporalis muscles before and 1, 3, and 6 months after the injection (Table 1).

In the temporalis muscles, CT showed a volume reduction of 14% over 6 months. The maximum reduction was observed at 3 months after the injection (Figure 2). Although the effect was gradually diminished, it was maintained until 6 months after the injection.

In the parotid glands, a total volume reduction of 6% was observed over 6 months, which was confirmed by clinical photographs (Figure 2) and CT measurements (Figure 3). After the procedure, the width of the face was decreased, and the jawline appeared narrow when viewed from the side. The largest reduction in the volume of the parotid glands was observed at 6 months after the injection. The patient was very pleased with the treatment outcome and did not complain of adverse effects such as dry mouth, decreased saliva production, and muscle weakness.

Patient 2

A healthy 35-year-old woman received PRA injection into both temporalis muscles (32 units per side). The changes in the volume of the temporalis muscles before and 1, 3, and 6 months after the injection were evaluated by CT. Over time, the volume of the right temporalis muscle was 29.65, 26.29, 24.15*, and 27.38 cm^3, whereas the volume of the left temporalis muscle was 31.21, 29.16, 25.79*, and 28.84 cm^3 (*largest volume reduction). The largest volume reduction during the study period was 19% in the right and 17% in the left temporalis muscle. Similar to the first patient, the reduction was maximal at 3 months after the injection (Figure 4). The patient also did not experience any adverse effects.

Discussion

As BTX-A injection into the temporalis muscles is usually performed for the treatment of headaches and bruxism, there are limited reports on BTX-A injection into only the temporalis muscles for cosmetic purposes.[4] We used PRA, which is considered to be equivalent to onabotulinum toxin-A (ONA, Botox, Allergan, Irvine, CA) with 1:1.[5]

To the best of our knowledge, this study is the first to show volume changes (measured with CT) in the temporalis muscles over time after BTX-A injection. Botulinum toxin A injection into the temporalis muscles could be an effective sole treatment that reduces the width of the upper face and makes the face look smaller. As patients with bruxism may

TABLE 1. Volume Changes in the Parotid Glands and Temporalis Muscles of Patient 1 Measured by Computed Tomography

	Temporalis Muscles		*Parotid Glands*	
Volume (cm^3)	*Right*	*Left*	*Right*	*Left*
Baseline	48.21	48.22	37.89	37.98
1 mo	45.27	44.66	36.39	36.85
3 mo	41.84*	42.88*	36.05	36.05
6 mo	46.44	47.20	35.85*	35.94*

*Largest reduction in volume during the study period.

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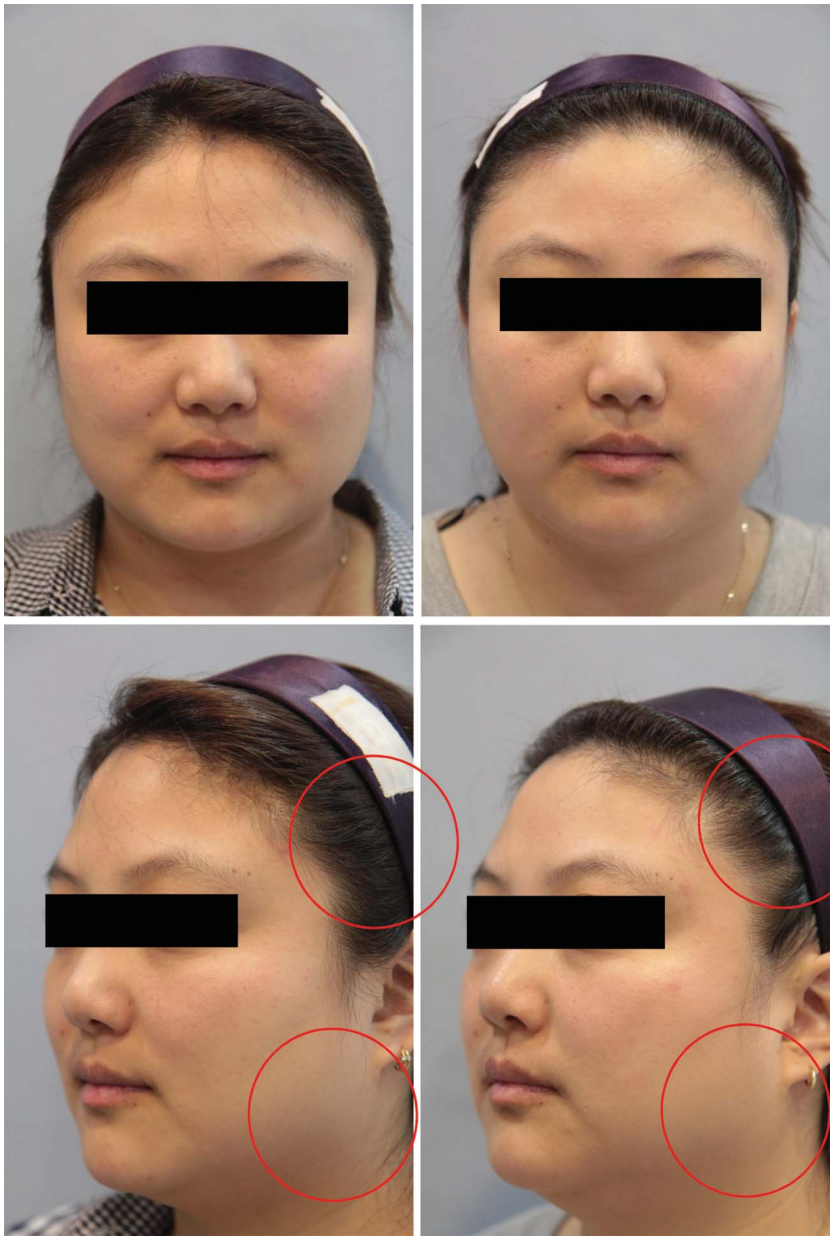


Figure 2. Clinical photographs before (top and bottom left) and 3 months after (top and bottom right) prabotulinum toxin Type A injection into the temporalis muscles and parotid glands.

have enlarged masseter and temporalis muscles, their facial width may be larger. Therefore, for these patients, BTX-A injection into the temporalis muscles can provide not only functional effects but also cosmetic effects.

Botulinum toxin A injections into the parotid glands to reduce the facial width have been reported previously.[4] However, to the best of our knowledge, there has been no report on the exact amount and duration of the effect in terms of parotid gland volume reduction over time after BTX injection. Previous studies are important in demonstrating that BTX-A injections into the masseter muscles and parotid glands are effective in reducing the lower facial width.[2–4] On the other hand, this study is significant because it shows quantitative changes over time based on CT measurements.

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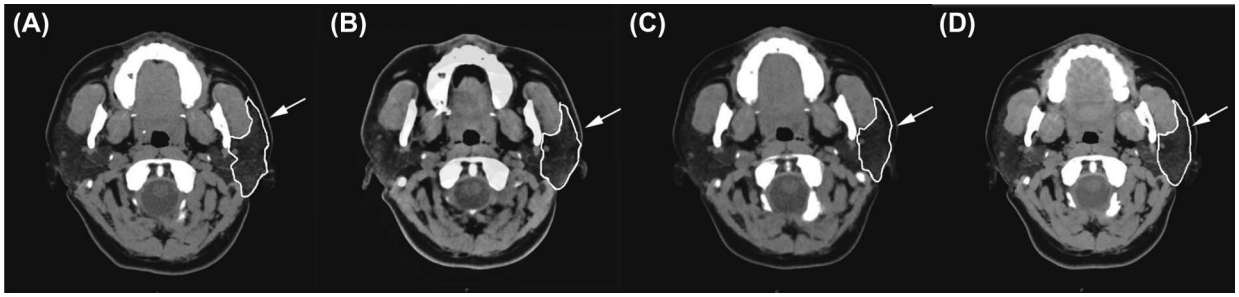


Figure 3. Volume changes in the parotid glands over time measured by computed tomography: (A) baseline, (B) 1 month, (C) 3 months, and (D) 6 months after the injection. Arrows indicate the area of the parotid gland.

The effect was maximal at 3 months after the injection into the temporalis muscles, which is consistent with the results of a previous study on BTX-A injection into the masseter muscles.[2] Interestingly, unlike the results for the muscles, the reduction in the volume of the parotid glands was maintained until 6 months after the injection. This suggests that there may be a difference in the muscle and gland responses after BTX-A injection. As muscles generally perform functions independent of each other, it is likely that atrophied muscles will be used continuously and will gradually recover. However, the mechanism of the salivary glands is somewhat different. Although the parotid glands are the largest of the salivary glands, other salivary glands including the submandibular, sublingual, and minor salivary glands can replace the functions of an atrophic parotid gland. In other words, the recovery of the parotid glands is slower because they are used less than the muscles after atrophy.

Compensatory hypertrophy can also occur in salivary glands. However, when we used 32 units of BTX-A per parotid gland, total volume reduction after the injection was only about 6%. Moreover, approximately 67% to 70% of saliva in the oral cavity is known to be produced by submandibular glands. Therefore, when 32 units of toxin per parotid gland were used, it is considered that

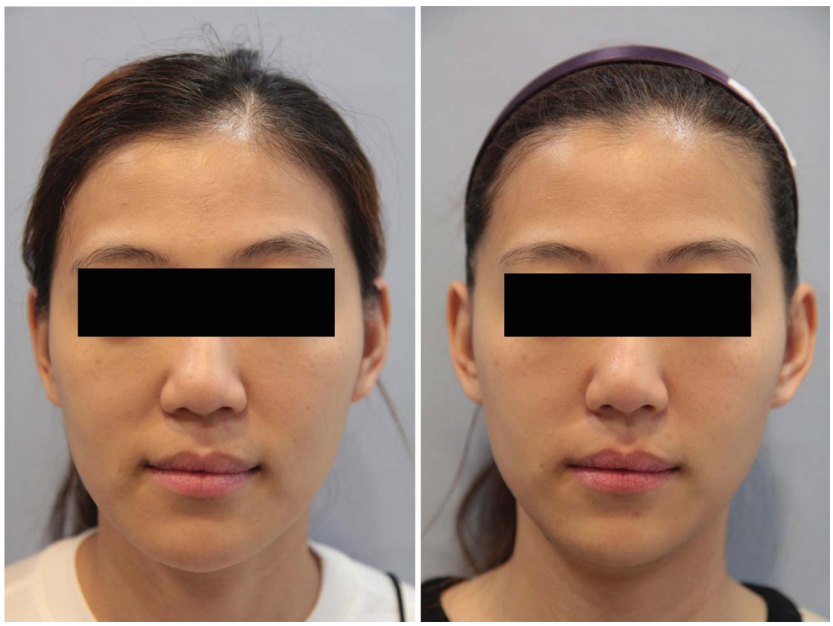


Figure 4. Clinical photographs before (left) and 3 months after (right) the prabotulinum toxin Type A injection into the temporalis muscles.

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compensatory hypertrophy of other salivary glands and oral dryness did not occur.[3,4]

In conclusion, PRA injection into the temporalis muscles and parotid glands may be a safe and novel treatment option for facial contouring.

Acknowledgments Patients provided written consent for the use of their images.

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